TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Topic – Practice and Revision

Lecture No.- 03

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Topics to be Covered

Topic

1:- Practice & Revision Set

MSB

1 → -ve
0 → +ve

$\boxed{1\,|\,1001}$ ← Binary number

1. Any signed negative binary number is recognised by its _________
   a) MSB   Most Significant Bit
   b) LSB   Least Significant Bit
   c) Byte   8 bits
   d) Nibble2   4 bits

2. If the decimal number is a fraction then its binary equivalent is obtained by _________ the number continuously by 2.
   a) Dividing
   b) Multiplying
   c) Adding
   d) Subtracting

$(24.48)_{10} \longrightarrow (\quad)_2$

$2 \,|\, 24$
$2 \,|\, 12 \;\; 0$
$2 \,|\, 6 \;\; 0$
$2 \,|\, 3 \;\; 0$
$\;\; 1 \;\; 1$

divide

$2 \times 0.48 = 0.96 \rightarrow 0$
$2 \times 0.96 =$

Topic- Practice

1. Answer: a
   Explanation: Any negative number is recognized by its MSB (Most Significant Bit). If it's 1, then ít's negative, else if it's 0, then positive.

2. Answer: b
   Explanation: On multiplying the decimal number continuously by 2, the binary equivalent is obtained by the collection of the integer part. However, if it's an integer, then it's binary equivalent is determined by dividing the number by 2 and collecting the remainders.

3. The representation of octal number (532.2)8 in decimal is ________

a) (346.25)10

b) (532.864)10

c) (340.67)10

d) (531.668)10

$(532.2)_8$ 10

$5\times8^2+3\times8^1+2\times8^0+2\times8^{-1}$

$320+24+2+0.25$

4. The decimal equivalent of the binary number (1011.011)2 is ________

a) (11.375)10

b) (10.123)10

c) (11.175)10

d) (9.23)10

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3. Answer: a

Explanation: Octal to Decimal conversion is obtained by multiplying 8 to the power of base index along with the value at that index position.

$(532.2)8 = 5 * 8^2 + 3 * 8^1 + 2 * 8^0 + 2 * 8^{-1} = (346.25)10$

4. Answer: a

Explanation: Binary to Decimal conversion is obtained by multiplying 2 to the power of base index along with the value at that index position.

$1 * 2^3 + 0 * 2^2 + 1 * 2^1 +1*2^0 + 0 * 2^{-1} +1 * 2^{-2} + 1 * 2^{-3} = (11.375)10$

Hence, (1011.011)2 = (11.375)10

3. Answer: a

Explanation: Octal to Decimal conversion is obtained by multiplying 8 to the power of base index along with the value at that index position.

$(532.2)8 = 5 * 8^2 + 3 * 8^1 + 2 * 8^0 + 2 * 8^{-1} = (346.25)10$

4. Answer: a

Explanation: Binary to Decimal conversion is obtained by multiplying 2 to the power of base index along with the value at that index position.

$1 * 2^3 + 0 * 2^2 + 1 * 2^1 +1*2^0 + 0 * 2^{-1} +1 * 2^{-2} + 1 * 2^{-3} = (11.375)10$

Hence, $(1011.011)2 = (11.375)10$

5. An important drawback of binary system is ________
a) It requires very large string of 1's and 0's to represent a decimal number
b) It requires sparingly small string of 1's and 0's to represent a decimal number
c) It requires large string of 1's and small string of 0's to represent a decimal number
d) It requires small string of 1's and large string of 0's to represent a decimal number

6. The largest two digit hexadecimal number is ________
a) (FE)16
b) (FD)16
c) (FF)16
d) (EF)16

$(FF)_{16}$

5 Answer: a

Explanation: The most vital drawback of binary system is that it requires very large string of 1's and 0's to represent a decimal number. Hence, Hexadecimal systems are used by processors for calculation purposes as it compresses the long binary strings into small parts.

6. Answer: c

Explanation: (FE)16 is 254 in decimal system, while (FD)16 is 253. (EF)16 is 239 in decimal system. And, (FF)16 is 255. Thus, The largest two-digit hexadecimal number is (FF)16.

5 Answer: a

Explanation: The most vital drawback of binary system is that it requires very large string of 1's and 0's to represent a decimal number. Hence, Hexadecimal systems are used by processors for calculation purposes as it compresses the long binary strings into small parts.

6. Answer: c

Explanation: (FE)16 is 254 in decimal system, while (FD)16 is 253. (EF)16 is 239 in decimal system. And, (FF)16 is 255. Thus, The largest two-digit hexadecimal number is (FF)16.

7. Perform binary addition: 101101 + 011011 = ?
a) 011010
b) 1010100
c) 101110
d) 1001000

$$\begin{array}{r} {}^{1\,1\,1\,1\,1\,} \\ 101101 \\ 011011 \\ \hline 1001000 \\ \hline \end{array}$$

8. Perform binary subtraction: 101111 – 010101 = ?
a) 100100
b) 010101
c) 011010
d) 011001

$$\begin{array}{r} 101111 \\ -\,010101 \\ \hline 011010 \\ \hline \end{array}$$

7. Perform binary addition: 101101 + 011011 = ?
a) 011010
b) 1010100
c) 101110
d) 1001000

8. Perform binary subtraction: 101111 – 010101 = ?
a) 100100
b) 010101
c) 011010
d) 011001

9. Perform multiplication of the binary numbers: 01001 × 01011 = ?

a) 001100011

b) 110011100

c) 010100110

d) 101010111

$$
\begin{array}{r}
01001 \\
\times\ 01011 \\
\hline
01001 \\
01001\times \\
00000\times\times \\
01001\times\times\times \\
00000\times\times\times\times \\
\hline
001100011 \\
\hline
\end{array}
$$

10. Divide the binary numbers: 111101 ÷ 1001 and find the remainder.

a) 0010

b) 1010

c) 1100

d) 0111

9. Perform multiplication of the binary numbers: 01001 × 01011 = ?
a) 001100011
b) 110011100
c) 010100110
d) 101010111

10. Divide the binary numbers: 111101 ÷ 1001 and find the remainder.
a) 0010
b) 1010
c) 1100
d) 0111

9

```
        0 1 0 0 1
      x 0 1 0 1 1
      -----------
        0 1 0 0 1
      0 1 0 0 1 0
    0 0 0 0 0 0 0
  0 1 0 0 1 0 0 0
0 0 0 0 0 0 0 0 0
-----------------
0 0 1 1 0 0 0 1 1
-----------------
```

10.

```
1 0 0 1 ) 1 1 1 1 0 1 ( 1 1
          1 0 0 1
          -------
          0 1 1 0 0
            1 0 0 1
          ---------
            0 1 1 1
```

```
1001) 111101 ( 11
      1001
      ----
      x1100
       1001
      -----
      00111  Remainder
```

11. 1's complement of 1011101 is ______________

a) 0101110

b) 1001101

c) 0100010

d) 1100101

0100010

12. 2's complement of 11001011 is ______________

a) 01010111

b) 11010100

c) 00110101

d) 11100010

00110100

+1

00110101

11. Answer: c
Explanation: 1's complement of a binary number is obtained by reversing the binary bits. All the 1's to 0's and 0's to 1's.

Thus, 1's complement of 1011101 = 0100010.

12. Answer: c
Explanation: 2's complement of a binary number is obtained by finding the 1's complement of the number and then adding 1 to it.
2's complement of 11001011 = 00110100 + 1 = 00110101.

13. On subtracting (010110)2 from (1011001)2 using 2's complement, we get ____________

a) 0111001
b) 1100101
c) 0110110
d) 1000011

14. On subtracting (001100)2 from (101001)2 using 2's complement, we get ____________

a) 1101100
b) 011101
c) 11010101
d) 11010111

13. On subtracting (010110)2 from (1011001)2 using 2's complement, we get ______________

a) 0111001
b) 1100101
c) 0110110
d) 1000011

101001
+1
101010

0000011
= 11

1011001 → Minuend
+101010 → Subtrahend (2's)
1 0000011

14. On subtracting (001100)2 from (101001)2 using 2's complement, we get ______________

a) 1101100
b) 011101
c) 11010101
d) 11010111

110011
+1
110100

101001 → Minuend
+110100
1 011101

101001
+1
101010

13

```
1's complement of subtrahend -        1 1 0 1 0 0 1
                                      _____________
                                      1 1 1
                     Minuend -        1 0 1 1 0 0 1
2's complement of subtrahend -        1 1 0 1 0 1 0
                                      _____________

                  Carry over -  1     1 0 0 0 0 1 1

Answer: 1000011
```

14

```
1's complement of subtrahend -        1 1 0 0 1 1
                                      ___________
                     Minuend -        1 0 1 0 0 1
2's complement of subtrahend -        1 1 0 1 0 0
                                      ___________
                  Carry over -  1     0 1 1 1 0 1

Answer: 011101
```

x 1 1 0 1 0 1
0 0 1 0 1 0
+1
0 0 1 0 1 1

Answer: d
Explanation: Steps For Subtraction using 2's complement are:
-> 2's complement of the subtrahend is determined and added to the minuend.
-> If the result has a carry, then it is dropped and the result is positive.
-> Else, if there is no carry, then 2's complement of the result is found out and a '-' sign preceeds the result.

14

```
1's complement of subtrahend -              1 1 0 0 1 1
                                           _____________
                 Minuend -                  1 0 1 0 0 1
2's complement of subtrahend -              1 1 0 1 0 0
                                           _____________
              Carry over -     1            0 1 1 1 0 1

Answer: 011101
```

Answer: b

Explanation: Steps For Subtraction using 2's complement are:

-> 2's complement of the subtrahend is determined and added to the minuend.

-> If the result has a carry, then it is dropped and the result is positive.

-> Else, if there is no carry, then 2's complement of the result is found out and a '-' sign preceeds the result.

15. The canonical sum of product form of the function y(A,B) = A + B is ___________

a) AB + BB + A'A

b) AB + AB' + A'B

c) BA + BA' + A'B'

d) AB' + A'B + A'B'

$$Y = A + B$$
$$= A(B+B') + B(A+A')$$
$$= AB + AB' + BA + BA'$$
$$= AB + AB' + BA'$$

$$A + A = A$$
$$AB + AB = AB$$

$(S_3) \cdot (S_2)(S_1)$

16.The expression Y=(A+B)(B+C)(C+A) shows the ___________ operation.

a) AND

b) POS

c) SOP

d) NAND

POS = Product of Sum

15. The canonical sum of product form of the function $y(A,B) = A + B$ is __________

a) $AB + BB + A'A$

b) $AB + AB' + A'B$

c) $BA + BA' + A'B'$

d) $AB' + A'B + A'B'$

16.The expression $Y=(A+B)(B+C)(C+A)$ shows the _________ operation.

a) AND

b) POS

c) SOP

d) NAND

15. Answer: b

Explanation: $A + B = A.1 + B.1 = A(B + B') + B(A + A') = AB + AB' + BA + BA' = AB + AB' + A'B$
$= AB + AB' + A'B$.

16. Answer: b

Explanation: The given expression has the operation sum as well as the product of that. So, it shows POS(product of sum) operation. SOP will be the sum of product terms

17. There are _______ cells in a 4-variable K-map.
a) 12
b) 16
c) 18
d) 8

No of Cell $= 2^n = 2^4 = 16$

18. Don't care conditions can be used for simplifying Boolean expressions in _____________
a) Registers
b) Terms
c) K-maps
d) Latches

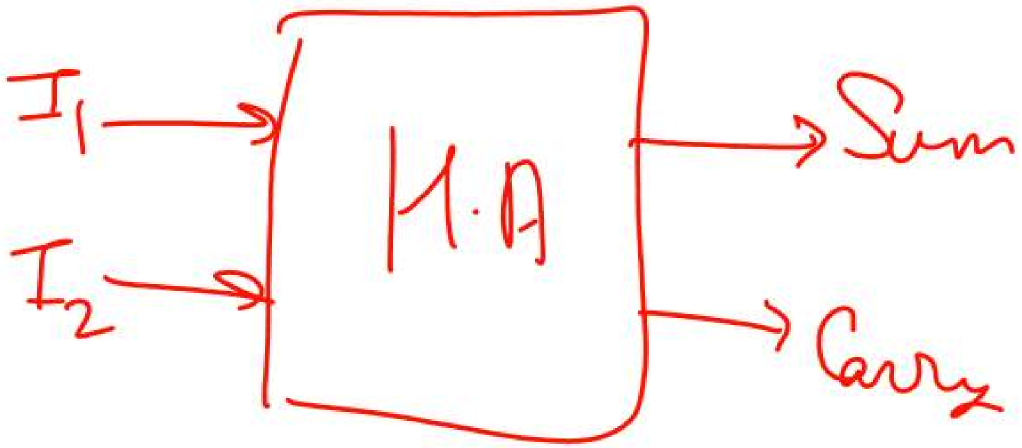
Topic- Practice

19. In a combinational circuit, the output at any time depends only on the ________ at that time.

a) Voltage
b) Intermediate values
c) Input values
d) Clock pulses

20. Total number of inputs in a half adder is ____________

a) 2
b) 3
c) 4
d) 1



21. If A and B are the inputs of a half adder, the sum is given by __________
a) A AND B
b) A OR B
c) A XOR B
d) A EX-NOR B

22. Half-adders have a major limitation in that they cannot __________
a) Accept a carry bit from a present stage
b) Accept a carry bit from a next stage
c) Accept a carry bit from a previous stage
d) Accept a carry bit from the following stages

21. Answer: c
Explanation: If A and B are the inputs of a half adder, the sum is given by A XOR B, while the carry is given by A AND B.

22. Answer: c
Explanation: Half-adders have a major limitation in that they cannot accept a carry bit from a previous stage, meaning that they cannot be chained together to add multi-bit numbers. However, the two output bits of a half-adder can also represent the result A+B=3 as sum and carry both being high.

23. How many AND, OR and EXOR gates are required for the configuration of full adder?
a) 1, 2, 2
b) 2, 1, 2
c) 3, 1, 2
d) 4, 0, 1

24. Full subtractor is used to perform subtraction of ___________
a) 2 bits
b) 3 bits
c) 4 bits
d) 8 bits

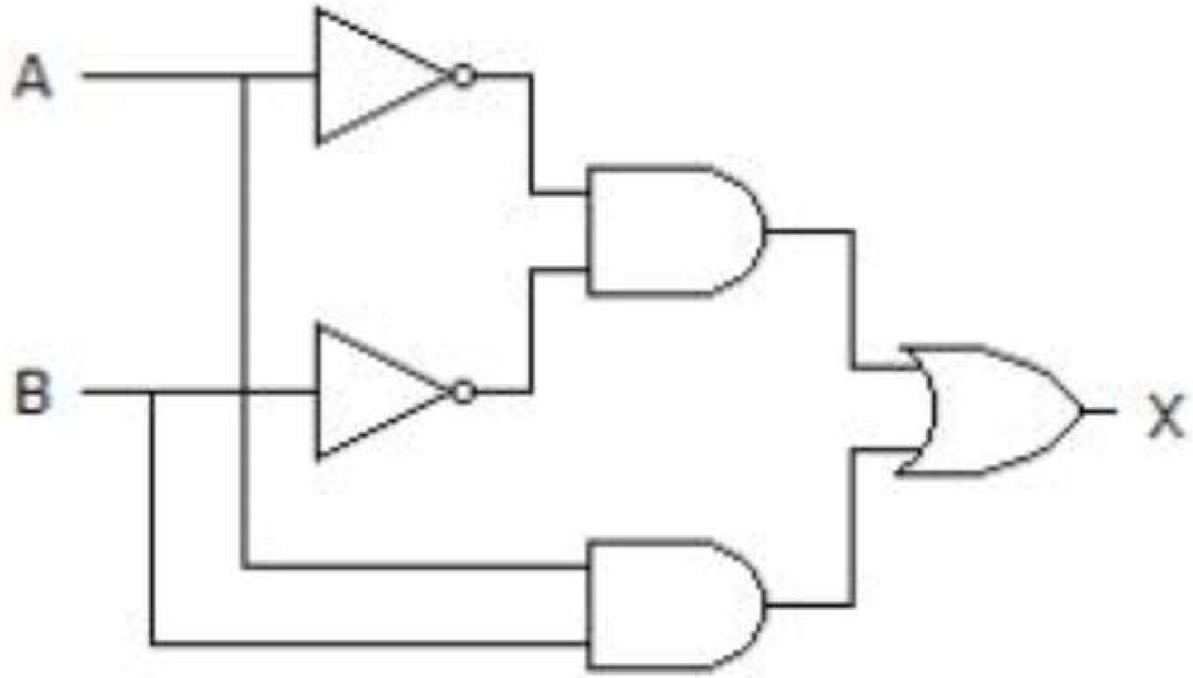
23. Answer: b

Explanation: There are 2 AND, 1 OR and 2 EXOR gates required for the configuration of full adder, provided using half adder. Otherwise, configuration of full adder would require 3 AND, 2 OR and 2 EXOR.

24. Answer: b

Explanation: Full subtractor is used to perform subtraction of 3 bits, namely minuend bit, subtrahend bit and borrow from the previous stage. However, it also produces 2 outputs BORROW and DIFFERENCE..

2. Which of the following logic expressions represents the logic diagram shown?



a) X=AB'+A'B

b) X=(AB)'+AB

c) X=(AB)'+A'B'

d) X=A'B'+AB

25. Answer: d

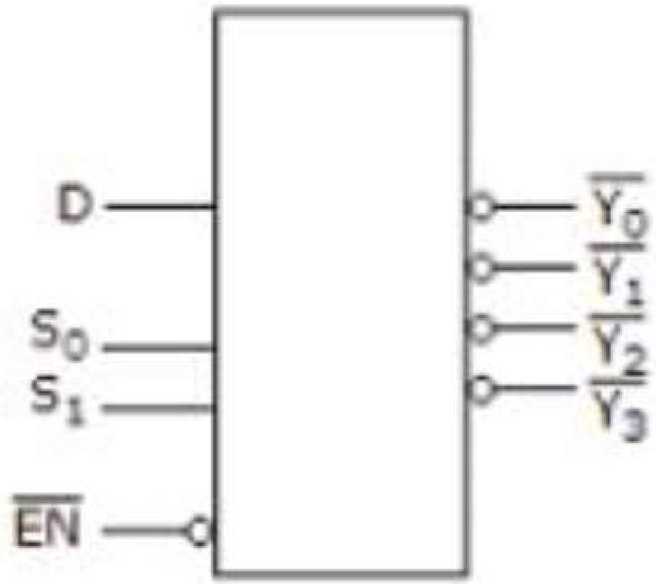
Explanation: 1st output of AND gate is = A'B'

2nd AND gate's output is = AB and,

OR gate's output is = (A'B')+(AB) = AB + A'B'.

26.

3. The device shown here is most likely a __



a) Comparator

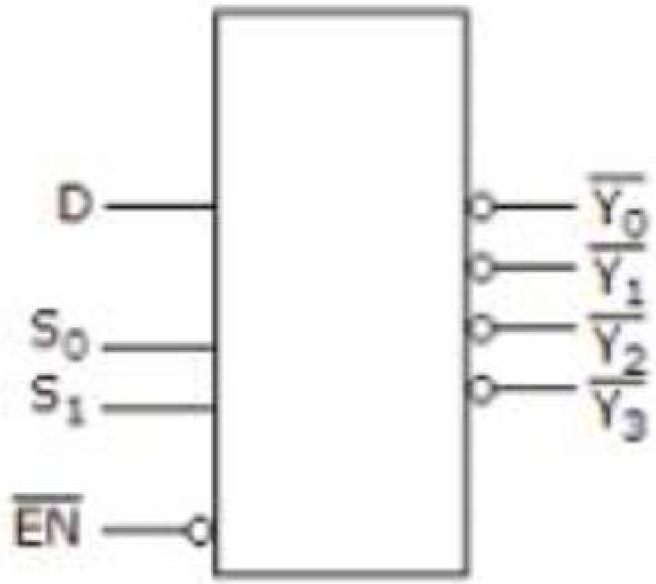
b) Multiplexer

c) Inverter

d) Demultiplexer

26.

3. The device shown here is most likely a __



a) Comparator

b) Multiplexer

c) Inverter

d) Demultiplexer

26. Answer: d

Explanation: The given diagram is demultiplexer, because it takes single input & gives many outputs. A demultiplexer is a combinational circuit that takes a single output and latches it to multiple outputs depending on the select lines.

27. What is data routing in a multiplexer?
    a) It spreads the information to the control unit
    b) It can be used to route data from one of several source to destination
    c) It is an application of multiplexer
    d) It can be used to route data and it is an application of multiplexer

28 .Which of the following circuit can be used as parallel to serial converter?
a) Multiplexer
b) Demultiplexer
c) Decoder
d) Digital counter

27. What is data routing in a multiplexer?
    a) It spreads the information to the control unit
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    c) It is an application of multiplexer
    d) It can be used to route data and it is an application of multiplexer

28 .Which of the following circuit can be used as parallel to serial converter?
a) Multiplexer
b) Demultiplexer
c) Decoder
d) Digital counter

27. Answer: d

Explanation: Multiplexing means passing more than one data through the same channel. Data routing is an application of multiplexer and it can be used to route data from one of several source to destination.

28 . Answer: a

Explanation: A combinational circuit that selects one from many inputs is known as Multiplexer. In multiplexer, different inputs are inserted parallely and then it gives one output which is in serial form.

29. Can an encoder be a transducer?
    a) Yes
    b) No
    c) May or may not be
    d) Both are not even related slightly

30 If two inputs are active on a priority encoder, which will be coded on the output?
a) The higher value
b) The lower value
c) Neither of the inputs
d) Both of the inputs

29. Can an encoder be a transducer?
a) Yes
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d) Both are not even related slightly

30 If two inputs are active on a priority encoder, which will be coded on the output?
a) The higher value
b) The lower value
c) Neither of the inputs
d) Both of the inputs

29. Answer: a

Explanation: Of course, a transducer is a device that has the capability to emit data as well as to accept. Transducer converts signal from one form of energy to another.

30 Answer: a

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. If two inputs are active on a priority encoder, the input of higher value will be coded in the output

THANK YOU